

Heterocyclic amines: Mutagens/carcinogens produced during cooking of meat and fish.

Research leading to the discovery of a series of mutagenic and carcinogenic heterocyclic amines (HCAs) was inspired by the idea that smoke produced during cooking of food, especially meat or fish, might be carcinogenic. More than ten kinds of HCAs, actually produced by cooking or heating of meat or fish, have now been isolated and their structures determined, most being previously unregistered compounds. They are highly mutagenic towards *Salmonella typhimurium* in the presence of S9 mix and are also mutagenic in vitro and in vivo toward mammalian cells. HCAs have now been chemically synthesized in quantity and subjected to long-term animal testing. When HCAs were fed in the diet, rodents developed cancers in many organs, including the colon, breast and prostate, and one HCA produced hepatomas in monkeys. The lesions exhibited alteration in genes including Apc, beta-catenin and Ha-ras, and these changes provide clues to the induction mechanisms. The HCAs are oxidized to hydroxyamino derivatives by cytochrome P450s, and further converted to ester forms by acetyltransferase and sulfotransferase. Eventually, they produce DNA adducts through the formation of N-C bonds at guanine bases. There are HCA-sensitive and resistant strains of rodents and a search for the responsible genes is now under way. While the content of HCAs in dishes consumed in ordinary life is low and not sufficient in itself to explain human cancer, the coexistence of many other mutagens/carcinogens of either autobiotic or xenobiotic type and the possibility that HCAs induce genomic instability and heightened sensitivity to tumor promoters suggest that avoidance of exposure to HCAs or reduction of HCAs' biological effects as far as possible are to be highly recommended. Usage of microwave ovens for cooking and supplementation of the diet, for example with soy-isoflavones, which have been found to suppress the occurrence of HCA-induced breast cancers, should be encouraged. Advice to the general public about how to reduce the carcinogenic load imposed by HCAs would be an important contribution to cancer prevention.